Band-1 Hadamard Order 4516 — Miyamoto (Double Paley-I seed H_1128); Gap 2052 Still OPEN

H_564/H_1128/C_1130/H_4516 PASS; 4516 Miyamoto-only bonus (NOT in SPK); gap 2052 still OPEN; conjecture OPEN; GATE-04 LOCKED

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2026-09-15

Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parents. `2026-09-15_Band1_Hadamard_4468_Miyamoto`; `2026-09-15_Band1_Hadamard_4388_Miyamoto` (Double Paley-I seed pattern).
Lab. `/workspace/experiments/band1/hadamard_4516_miyamoto/`.
Cutoff. 2026-09-15T16:40:29Z (~12:40PM ET).

Honesty stamp (non-negotiable).
This memo claims **finite constructions of orders 564, 1128 and 4516 only** (verified $\max|HH^\top - nI| = 0$), via Miyamoto with Double(Paley-I) seed H_{1128} (H_{564} = Paley-I($q = 563$); H_{1128} = Sylvester-double(H_{564})). Paley-II path for H_{1128} is **FAIL** ($q = 563 \equiv 3 \pmod 4$; needs $\equiv 1$). Paley-I direct for H_{1128} is **SKIP** ($q = 1127 = 7^2 \cdot 23$ not prime). It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. Standard constructive gap **2052** remains **OPEN** under Sylvester∪Paley∪Kronecker (SPK) closure. Order 4516 is a **Miyamoto-only bonus (NOT in SPK)**: Paley-I **FAIL** ($q = 4515 = 3 \cdot 5 \cdot 7 \cdot 43$ not prime); Paley-II **FAIL** ($q = 2257 = 37 \cdot 61$ not prime); Kronecker **FAIL** ($4516 = 4 \cdot 1129$, 1129 not Hadamard-admissible). `clay_closed = false`.

Abstract

Prior Band-1 ships constructed Miyamoto-eligible orders through 4468 on box. This increment ships **order 4516** ($4516 = 4 \cdot 1129$, 1129 prime $\equiv 1 \pmod 4$) via Miyamoto with a **Double(Paley-I) seed** H_{1128}. Symmetric Paley conference C_{1130} over prime 1129; $H_{4516} = \text{Miyamoto}(H_{1128}, C_{1130})$. Unit tests **44/44 PASS**. SPK next gap remains **2052** (OPEN). Order 4516 is **not** in SPK (Miyamoto-only bonus). The **full Hadamard conjecture remains OPEN**.

Keywords. Hadamard matrix; order 4516; Miyamoto; Paley type I; Sylvester double; Paley conference; constructive gap 2052; SPK; Band-1; Miyamoto-only bonus; non-claim.

1. Statement lock

Definition. A Hadamard matrix of order n is an $n \times n$ matrix H with entries in ± 1 satisfying $HH^\top = nI$.

Finite lemmas (this increment). 1. H_{564} constructed and verified (Paley type I, $q = 563$). 2. H_{1128} constructed and verified (Sylvester double of H_{564}). Paley-II path for H_{1128} **FAIL** ($q = 563 \equiv 3 \pmod 4$). Paley-I direct **SKIP** ($q = 1127 = 7^2 \cdot 23$ not prime). 3. C_{1130} constructed and verified (Paley conference, $q = 1129$): $CC^\top = 1129I$, zero diagonal. 4. H_{4516} constructed and verified (Miyamoto). **Not** the Hadamard conjecture. 5. SPK: $4516 \notin$ SPK (Miyamoto-only). Paley-I/II/Kronecker paths for 4516 **rejected**. Next constructive gap remains **2052** (OPEN). Verified H_{564} and H_{1128} enter SPK as classical Double Paley-I seeds.

2. Arithmetic and eligibility

Quantity	Value
Target order	$4516 = 4 \cdot 1129$
$q = 1129$	prime, $1129 \equiv 1 \pmod 4 \rightarrow$ Miyamoto-eligible
Needed H_{q-1}	H_{1128}
Needed conference	order $1130 = q + 1$
H_{1128} path (used)	Double(Paley-I): $H_{564} = \text{Paley-I}(q = 563)$; $H_{1128} = \text{Sylvester-double}(H_{564})$
H_{1128} Paley-II	**FAIL**: $2(q + 1) = 1128 \Rightarrow q = 563 \equiv 3 \pmod 4$ (needs $\equiv 1$)
H_{1128} Paley-I direct	**SKIP**: $q + 1 = 1128 \Rightarrow q = 1127 = 7^2 \cdot 23$ not prime
Conference path	Symmetric Paley over prime 1129
Paley-I for 4516	**REJECTED**: $q' = 4515 = 3 \cdot 5 \cdot 7 \cdot 43$ not prime
Paley-II for 4516	**REJECTED**: $q' = 2257 = 37 \cdot 61$ not prime

Quantity	Value
Kronecker for 4516	**REJECTED**: only 4×1129; $1129 \not\equiv 0 \pmod 4$ (not admissible)

3. Construction results

Object	Method	$\max \lvert HH^\top - nI \rvert$ (or conf)	Wall (s)	Status
H_{564}	Paley type I, $q = 563$	0	0.2582	**PASS**
H_{1128}	Sylvester double of H_{564}	0	0.6621	**PASS**
H_{1128} Paley-II	$q = 563 \equiv 3 \pmod 4$	—	—	**FAIL** (needs $\equiv 1$)
H_{1128} Paley-I direct	$q = 1127 = 7^2 \cdot 23$	—	—	**SKIP** (not prime)
C_{1130}	Paley conference, $q = 1129$	0	1.261	**PASS**
H_{4516}	Miyamoto(H_{1128}, C_{1130})	0	41.6142	**PASS**

Primary pipeline wall (incl. unit tests + SPK refresh) ≈ 45.1082 s. Compressed ± 1 matrices saved as `H564_pm1.npz`, `H1128_pm1.npz`, `C1130_pm1.npz`, `H4516_pm1.npz` (int8).

4. SPK closure note

Seeds: all admissible ≤ 2000 $\cup$ on-box prior classical/SPK ships (through 4036/3908/3748/… + **1020/1116** Paley-II + **1032/1092** Paley-I + **524/1048/548/1096** Double Paley-I; **exclude** 4468/4388/4372/4196/4132/4084/3716/3524/3284/3076/3044/2836/2692/2564 Miyamoto-only) $\cup$ verified H_{564}/H_{1128} (Double Paley-I) $\cup$ Sylvester $\cup$ Paley I/II, closed under Kronecker (limit 4600). **Miyamoto is not folded into SPK**; order 4516 remains a Miyamoto-only bonus.

Quantity	Value
Known count	983
4516 in SPK	**False** (Miyamoto-only)
1128 in SPK	**True** (Double Paley-I seed)
564 in SPK	**True** (Paley-I seed)
1116 in SPK	**True** (prior Paley-II)

Quantity	Value
Next constructive gap	**2052** (OPEN)

5. Gap 2052 (method triage / non-finding ONLY)

Paley-I, Paley-II, Kronecker, and Miyamoto-with-Paley-conference paths for order 2052 all **FAIL** on-box arithmetic. This is a **non-finding**, not a counterexample. Gap **2052 remains OPEN**. Conjecture **OPEN**.

6. Non-claims

- No full Hadamard conjecture claim.
- `clay_closed = false` forever in this ship.
- GATE-04 collection **LOCKED**; no invented J.
- No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / finished-ToE claims.
- Order 4516 is a **Miyamoto-only bonus (NOT in SPK)**.

7. Next optional

- q=1153 → n=4612 via PaleyI(q=1151) (or Kronecker notes)
- q=1229 → n=4916 via PaleyII(q=613)
- method-attack SPK gap **2052**
- Track B privacy / Object 1 meter await

8. Reproducibility

```
cd /workspace/experiments/band1/hadamard_4516_miyamoto
python3 verify_hadamard_4516.py
```

Artifacts: `results.json`, `literature_ledger.json`, `SUMMARY.txt`, `*.npz`, this PAPER.